MAANAS TYAGI

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PROFESSIONAL SUMMARY

AI engineer and open-source maintainer with 100+ merged pull requests, including Hugging Face, and two AI products built from 0 to 1 in production. Focused on LLM applications, RAG evaluation, and agentic systems. Author of one preprint and one paper under review on LLM behavior.

SELECTED IMPACT

- Co-built TrenTorch, an open-source ML education platform: 750+ stars, 6,000+ users, 7,000+ unique clones, 50,000+ organic X views
- 100+ merged PRs in open source, including 65+ to TrenTorch, each reviewed and accepted by maintainers
- Open-Source Contributor to Hugging Face (tau, Transformers): merged 4 PRs, including the NVIDIA NIM model provider, a Gemini tool-call (thoughtSignature) fix, a Windows CLI UTF-8 crash fix, and a TrainingArguments docs fix
- Built two products from 0 to 1 at Sstudize (StudentGPT, Sstudi Rating), including an AI tutor engineered to a $1 per student per year AI cost budget

PROFESSIONAL EXPERIENCE

AI Engineer & Consultant *October 2026 – Present*
Swayu · Remote

- Own an entire feature of the flagship product end to end, from architecture and AI integration through production delivery
- Build marketing and go-to-market (GTM) automation pipelines that replace manual outreach and campaign workflows with LLM-driven systems

Core Engineer & Co-Maintainer *August 2026 – Present*
TrenTorch (open source) · Remote · trentorch.com | github.com/TrenTorch/TrenTorch

- Co-maintain the platform (750+ stars, 6,000+ users) with 65+ merged PRs across the tren CLI and TrenTorch-Web browser IDE; own the CLI: test runner, Jupyter integration, benchmarking, Windows compatibility
- Authored 25+ curriculum modules: quantization, mixed precision, CUDA kernels, RLHF/DPO/GRPO, deployment, and drift monitoring

LLM Engineer *July 2026 – November 2026*
Sstudize · Remote

- Built StudentGPT from 0 to 1, an AI tutor for JEE/NEET students: streaming FastAPI backend with five tutoring modes, pgvector semantic cache, OCR image input, and daily credits; merged to the platform repo with CI and 22 passing tests
- Engineered AI cost to a $1 per student per year budget: ~$0.0001 per message (~10,400 messages per student-year) using one low-cost model and OCR instead of vision, avoiding a 16-20x cost per image query
- Built Sstudi Rating from 0 to 1, an Elo-style skill rating engine updated on every test submission; shipped to production and verified on 540 real questions with 0 calculation errors against hand-computed values

AI & Analytics Intern (*Forward Deployed Engineer Level Work*) *May 2026 – June 2026*
Genpact · Jaipur, India

- Replaced a $3,000 licensed reconciliation tool with a Python pipeline: 10,000-row files from 30+ minutes to under 5; used by tens of colleagues
- Built purchase, invoice, and GRNI reconciliation engines for Fortune 500 clients, replacing manual pivot-table work with single-file uploads

PUBLICATIONS

- **Instruction Fidelity Decay in Small Language Model Agents: Semantic Recall Dissociates from Format Compliance Under Semantic Interference** (Under Review)
- **Front-Loading Bias in Large Language Models: A Multi-Model Empirical Analysis** (ResearchGate Preprint) – doi.org/10.13140/RG.2.2.32504.92164

WRITING

- I got internship offer from my opensource work. Here's how you can do the same. (X article, Oct 2026, 6K views)
- I benchmarked Sarvam's Models on Eminem's Rapgod, just because I can. (X article, Oct 2026)
- How I would learn AI Engineering from scratch. (X article, Aug 2026, 4K views)

PROJECTS

RAG Eval System | *LangChain · Chroma · RAGAS · Python* *July 2026*
github.com/maanas1234/RAG-Eval-System

- Built a RAG evaluation harness (Precision/Recall/MRR, RAGAS faithfulness/relevancy) scored independently for retrieval and generation on 33 hand-labeled queries over a 69-document corpus
- Ablation study: semantic-only retrieval outperformed hybrid BM25+semantic retrieval on every metric (Recall@5: 0.848 vs. 0.818, Faithfulness: 0.788 vs. hybrid), overturning the assumed hybrid-is-better default

SLM Pretraining From Scratch | *PyTorch · Python* *June – July 2026*
github.com/maanas1234/small-llm-from-scratch

- Built a GPT-style decoder-only transformer from scratch (attention, tokenizer, training loop; 2.1M parameters), reaching validation loss 1.74 (~5.7 perplexity); ablated 5 configurations (120K to 4.8M parameters) on overfitting in a low-data regime

EDUCATION

Vellore Institute of Technology *Vellore, India*
Bachelor of Technology in Information Technology, 2024 – 2028

SKILLS

Languages: Python, SQL, C
RAG / Retrieval: LangChain, RAGAS, Chroma, FAISS, Embeddings, Vector Databases, Semantic Search, CRAG
Modeling: PyTorch, Machine Learning, Deep Learning, Transformer Architecture, Fine-Tuning (LoRA/PEFT), Prompt Engineering
Backend / Tools: Hugging Face, FastAPI, Flask, React, Pandas, NumPy, Docker, Git, GitHub Actions, CI/CD, PostgreSQL, MySQL